

ANANDA WUTWANNA

IoT & Embedded Systems Engineer

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SUMMARY

IoT and embedded systems engineer with hands-on experience designing and building firmware for microcontrollers (ESP32, Arduino) and the full-stack platforms that control them. Delivered production C/C++ firmware on ESP-IDF and FreeRTOS for industrial vertical-farming automation — rail-guided vehicle and lift motion control, ball-valve irrigation, and EC-based nutrient dosing — alongside the FastAPI/React control systems that orchestrate them over MQTT. Strong at taking a system from requirement through firmware, integration, and real-world deployment, with an emphasis on reliability, offline-tolerant operation, and maintainable code.

TECHNICAL SKILLS

- **Microcontrollers / MCUs:** ESP32, Arduino; RGV & lift motor control, valve control
- **Embedded OS / RTOS:** FreeRTOS, ESP-IDF, Linux (basic); NVS persistent config
- **Firmware (C/C++):** PWM pump/motor control, ISR-based pulse counting, OTA updates
- **Communication / Protocols:** MQTT, I2C, Modbus, RS-485, UART/Serial, ADC, TCP/UDP, WebSocket, REST
- **Sensors & control:** EC, flow sensing, PM2.5/PM10; PWM pump and valve control
- **Tooling:** PlatformIO, ESP-IDF, Arduino IDE, Visual Studio Code, Docker, Git/GitLab CI
- **Backend / Full-stack:** Python FastAPI, Node.js, Node-RED, React + TypeScript, PostgreSQL, MongoDB, MySQL
- **OS:** Linux (basic), Windows

WORK EXPERIENCE

Civic Agrotech — IoT Imagineer

Oct 2025 – Present

Design and develop the IoT and firmware stack for an automated Plant Factory with Artificial Lighting (PFAL) — a vertical-farming facility that moves and grows crop trays without manual handling.

- Developed the **automation firmware for Rail-Guided Vehicles (RGV) and multi-floor lifts** that transport crop trays through the facility — coordinating precise tray put-away, retrieval, and repositioning over MQTT.
- Built the **PFAL automation control platform** (React + FastAPI + MongoDB + MQTT) that unifies control and real-time monitoring of the RGV, lift, valve, and water systems.
- Wrote **ESP32 / FreeRTOS firmware for the ball-valve irrigation controllers**, driving multiple valves via I2C expanders and GPIO with MQTT control, Wi-Fi provisioning, and live status telemetry.
- Built the **water-refill and nutrient-dosing controller** (ESP32) with EC-based dosing, multi-point calibration, PWM pump control, and flow-sensor measurement.
- Authored technical documentation and grow-planning analysis for the facility.

Advisory — Software Analyst

2025 – Oct 2025

- Collaborated with PO/BA and UX/UI designers to evaluate technical feasibility of system requirements.
- Prepared sections of the SRS (swimlane diagrams, API documentation, ER diagrams); defined and analyzed system issues; designed and tested standards and solutions.
- Planned and estimated resources across the software development lifecycle; verified deliverables through system testing; provided user training and operational support.

ADC Microsystems · 2020 – 2025

Software Analyst

2022 – 2025

- Analyzed and designed systems from client requirements into practical solutions.
- Researched technical approaches to improve systems and meet user demands.

- Designed appropriate system architectures and coordinated with the development team to ensure system-design compliance.

Project Coordinator

2021 – 2022

- Coordinated a small development team across multiple concurrent projects.
- Analyzed and developed projects from client requirements into practical systems.
- Coordinated between development teams and clients to ensure on-time project delivery.

IoT Developer

2020 – 2021

- Developed and deployed IoT solutions from prototype to production.
- Oversaw development through deployment and installation phases.
- Conducted testing and troubleshooting in real operating environments.

KEY PROJECTS

PFAL — Vertical Farming Automation (Civic Agrotech)

End-to-end IoT automation for a Plant Factory with Artificial Lighting — an automated vertical-farming facility that moves and grows crop trays with no manual handling. Led the firmware and integration work that turns physical machinery into a coordinated, remotely controlled system.

- **Tray automation system (core):** orchestrates the full tray workflow — moving trays in and out of storage and repositioning them between slots — by coordinating the RGV and lift in sequence.
- **RGV (Rail-Guided Vehicle):** firmware to drive trays along the rails to the target position.
- **Lift:** firmware to raise and lower trays between floors.
- **Valve:** ESP32 / FreeRTOS controller for the ball-valve irrigation system (MQTT, Wi-Fi provisioning, live status telemetry).
- **Nutrient-mixing system:** ESP32 controller that mixes and dispenses nutrient solution with EC-based dosing and PWM pump control.
- **Control platform:** a React + FastAPI + MongoDB + MQTT application that unifies real-time control and monitoring of the RGV, lift, valve, and nutrient-mixing systems in one interface.

Water Bill Management System (ADC Microsystems)

Multi-platform utility billing for local government — POS field terminals, web admin portal, and Line OA consumer interface; RFID/NFC identification, OCR meter reading, real-time payments; PostgreSQL schema and workflow design across React Native / React JS / Line OA modules.

Air Quality Monitoring System (ADC Microsystems)

Real-time air-quality platform (React JS + Node.js + MySQL) with an MQTT IoT sensor network (PM2.5, PM10, temp, humidity), AQI calculation, Line OA alerts, and exportable reporting for authorities.

Beacon-based Public Announcement System (ADC Microsystems)

Location-aware BLE beacon system (React JS + Node.js + PostgreSQL) with Line Messaging API integration, role-based access control, and an engagement analytics dashboard.

Smart Watch Services (ADC Microsystems)

IoT healthcare backend ingesting real-time data from multiple smartwatch models (UDP/MQTT); RESTful + WebSocket APIs for heart rate, blood pressure, SPO2 and temperature; Node-RED pipeline; SOS/location tracking; PostgreSQL storage.

EDUCATION

Bachelor of Engineering in Computer Engineering — Naresuan University

Aug 2016 – Aug 2020